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MILITARY SPECIFICATION

RANGE FINDER, FIRE CONTROL: M24

1. SCOPE

1.1 This specification covers one type of coincidence image instrument known as Range Finder: M24, for determining range measurement in meters (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830 — Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of.

MIL-F-13926 — Fire Control Materiel, General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

F10511227 — Range Finder, Fire Control: M24.

LSQAP 10511227 — List of Supplementary Quality Assurance Provisions for Range Finder: M24.

PACKAGING DATA SHEET

MIL-P-14232/P10511227 — Packaging of Range Finder, Fire Control: M24.

(Copies of specifications, standards, drawings and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1240

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the drawings, general specifications and material specifications forming part of this specification.

3.2 Fabrication. The range finder shall be manufactured in accordance with Drawing F10511227 "Range Finder, Fire Control: M24" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General specifications. The contractor shall be responsible for adherence to, and compliance with the following requirements of Specifications MIL–O–13830 and MIL–F–13926.

3.3.1 *MIL–O–13830.*

(a) Cleanliness.

(b) Illuminated reticle.

3.3.2 *MIL–F–13926.*

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Electrical and electronic assemblies.

(e) Part identification and marking.

(f) Workmanship.

3.4 Pre-vibration.

3.4.1 *Orientation.* The range finder prior to vibration shall be set up in a horizontal position as shown on the Assembly View Drawing F10511227 and all instructions thereon, and the following conditions shall be established prior to testing.

(a) The line through the centers of the two mounting bearings horizontal.

(b) Identified production end housings in place.

(c) Level vials and reticles illuminated.

(d) Elevation and cant level bubbles centered within width of a vial graduation line.

(e) Occluder set at ranging position.

(f) Filters out of the optical line of sight.

(g) Cant correction knob in detent.

(h) A target provided at 1500 meters plus or minus 100 meters and the range scale set to agree with the range chosen.

(i) Set the ICS knob to within one division of its midpoint (zero) position of total travel, and set the halving knob to within ten degrees of its midpoint of total travel.

(j) Set the vertical calibration knob at midpoint of total travel within 10 degrees and slip the scale to align indices. Establish vertical coincidence of the target image.

(k) With the horizontal calibration knob set a midpoint of total travel within one division on the scale, make a series of five horizontal coincidence readings on the target image. Rotate the knob to the median reading.

(l) Set the deflection (azimuth) and elevation boresight knobs at midpoint of total travel within 0.1 mil graduation. Slip the deflection scales to the three (3) position and slip the elevation scales to the four (4) position.

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3.4.2 *Sighting reticle alignment.* The line of sight of the left telescope through the boresight mark (center of the cross of the sighting reticle) shall be horizontal within 0.5 mil and shall be 1.0 mil to the right of the perpendicular to a line through the center of the two mounting bearings within 0.5 mil. The line of sight of the right telescope through the boresight mark shall be such that its aiming point coincides with the aiming point of the left telescope within a maximum angular separation of 0.05 mil in azimuth and 0.5 mil in elevation without filters.

3.4.2.1 *Filters.* The introduction of filters into the optical system shall not cause a change in sighting reticle alignment in either optical system in excess of 0.1 mil in azimuth or elevation. Change in coincidence calibration shall not exceed more than one unit of error (UOE) horizontal calibration or 0.1 mil in vertical calibration.

3.4.3 *Collimation.* With the eyepieces set at infinity focus, the rays from the same point on the target emerging from the two eyepieces shall not diverge to less than 5 or more than 25 minutes of arc horizontally and shall not exceed 10 minutes of arc in dipvergence. These conditions shall be met with interpupillary knob set at any position between 58 and 72 millimeters at any elevation of the range finder.

3.4.4 *Knob travel.* (Calibration Knobs.)

3.4.4.1 *Horizontal.* The horizontal calibration knob shall read within 10 UOE of its midpoint of total travel when horizontal coincidence is established at 1500 meter plus or minus 100 meters.

3.4.4.2 *Vertical.* The vertical calibration knob shall be within 30 degrees of its previously established midpoint. The scale shall then be slipped to align the indices.

3.4.5 *Calibration reticle setting accuracy.*

3.4.5.1 *Horizontal.* The horizontal coincidence of the calibrating reticles shall be within 10 UOE. Five median readings shall be made with the (ICS) knob. The median of the base five readings shall read within 10 UOE of the midpoint (zero) position.

3.4.5.2 *Vertical.* The vertical coincidence of the calibrating reticle shall be within 45 degrees of the midpoint position of the halving knob established in 3.4.1 (j).

3.5 Vibration. The range finder shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds.

3.6 Post-vibration. With the range finder restored to the condition of 3.4.1 (a) to (l) inclusive the range finder shall again meet the requirements of 3.4.2 and 3.4.3 with the following exceptions:

(a) The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.20 mil in azimuth and 0.20 mil in elevation from the values established in 3.4.2. The line of sight of the right telescope through the boresight mark shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.05 mil.

(b) Collimation shall not have changed by more than 5 minutes in divergence and not more than 5 minutes in dipvergence and shall remain within the limitations of 3.4.3.

3.6.1 *Target image.* The target image shall not have changed coincidence from the actual value established prior to vibration by more than 0.5 mil in vertical calibration or more than 20 UOE in horizontal calibra-

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tion as measured against a calibrated external target using a median of five coincident settings made with the horizontal calibration knob.

3.6.2 *Coincident reticle setting.* With the horizontal calibration knob at the new median setting (3.6.1) and using the ISC knob, obtain a new median from 5 ICS settings of the calibrating reticle, the new median reading shall not differ by more than 5 UOE from the actual median reading established in 3.4.5.1 prior to vibration.

3.6.3 *Coincident reticle position.* With the boresight knobs positioned as in 3.6 (a), the bottom of the left coincidence reticle shall be 5 to 7 mils above (elevation) the center of the boresight cross and shall have the same azimuth setting within plus or minus one mil. With the target images in coincidence and the calibrating knobs, ICS and halving knobs set as in 3.6.1 and 3.6.2 the right coincidence reticle shall overlap the left coincidence reticle in elevation by at least one mil. The requirements of 3.6.3.1 and 3.6.3.2 shall be met with the range finder reset in accordance with 3.6.1.

3.6.3.1 *Vertical movement accuracy.* When the vertical calibration knob is rotated to its maximum counterclockwise (CCW) position and the reticle returned to its original position by means of the halving knob, the median of 5 deflection readings (UOE) of the calibrating reticle using the ICS knob shall not differ by more than 10 UOE from the median of 5 deflection readings obtained when the vertical calibration knob is rotated to its maximum clockwise (CW) position and the reticle to coincidence by means of the halving knob.

3.6.3.2 *Horizontal movement accuracy.* With the horizontal calibration knob rotated from the median setting (coincident position of left and right images) to plus 40 and minus 40 UOE, the range for each extreme shall be equivalent to the corresponding change in the number of UOE on the knob scale within 10 percent. In addition, 5 UOE are allowed to cover the error in the position of the horizontal calibrating wedge relative to the ranging plane and its angular tolerance.

3.6.4 *Dirt and bright spots.*

3.6.4.1 *Dirt.* The maximum real diameter of any one defect shall not exceed 0.2 mil in the outer 22½ degrees, and 0.05 mil in the central 17½ degrees of the apparent field as received at the eyepiece. The sum of the maximum real diameter of dirt and pinholes in the range finder shall not exceed 1.0 mil. The range finder shall meet the cleanliness requirements of Specification MIL–O–13830. Visible dirt on the optical surfaces shall not exceed in size and number the limitations for size and number of digs permitted for each surface as specified on applicable drawings. There shall be no other dirt or imperfections on other surfaces which interferes with ranging or observation or adversely affects image quality.

3.6.4.2 *Bright spots.* Bright spots (pinholes) in the reticle shall not exceed in size and number the limitations for size and number of digs permitted for that surface as specified on the applicable drawings.

3.6.5 *Parallax.*

3.6.5.1 *Parallax between sighting reticle and target images.* Utilizing a target at 1500 meter plus or minus 100 meters and with the range scale set to agree with the range chosen, parallax in the right and left optical system on the optical axis between the target and the boresight mark images shall not exceed 0.10 mil.

3.6.5.2 *Parallax of the calibrating reticles.* The median of five ICS readings taken with the eye centered at 0.030 inch to the left and

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the median of five readings taken with the eye centered at 0.030 inch to the right of the center of each exit pupil shall not differ by more than 6 UOE in the right and left optical systems as measured on the ICS knob.

3.6.5.3 *Parallax of the target images.* Utilizing a target at 1500 meter plus or minus 100 meters the median of five coincidence readings taken with the eye centered 0.030 inch to the left of the center of the exit pupil shall not differ from the median of five coincidence readings taken with the eye centered 0.030 inch to the right of the center of the exit pupil by more than 6 UOE in the right and left optical systems as read on the horizontal calibration knob.

3.6.6 *Eyepiece.*

3.6.6.1 *Interpupillary adjustment.* The interpupillary scale shall indicate the correct interpupillary distance within plus or minus 0.5 mm at any setting of the interpupillary scale.

3.6.6.2 *Eyepiece focus.* Eyepiece focus shall be adjustable over the range of plus and minus three diopters as read on the eyepiece scale.

3.6.6.3 *Diopter setting.* The zero of the diopter scale of each eyepiece shall indicate zero diopters within 0.25 diopters.

3.6.6.4 *Exit pupil diameter.* The minimum diameter of the exit pupils when viewed in the direction of the optical axis shall be 0.138 inch (3.5 mm).

3.6.6.5 *Exit pupil coincidence.* The lack of coincidence of the exit pupils in either eyepiece shall not exceed 0.30 millimeters.

3.6.6.6 *Magnification.* Magnification of the left and right systems shall be equal within one percent.

3.6.7 *Resolution.* Resolution on the optical axis of each optical system shall be six seconds of arc or less.

3.6.8 *Light transmission.* The difference in light transmission between the two optical systems shall not exceed twenty percent of the higher transmission value.

3.6.9 *Sighting reticle measurement.* The angular value of the total horizontal lines and spaces of the reticles in either field of view shall be forty mils plus or minus two mils.

3.6.10 *Parallelism of sighting reticle and image.* When the sighting reticle is projected into either eyepiece the longest horizontal reticle line (40 mil line) shall be parallel to the image of a horizontal target line within forty three minutes of arc. (Forty three minutes of arc is equivalent to an angle subtended by an 0.50 mil block at one extremity of the reticle line when the other extremity of the reticle line is in coincidence with the target line).

3.6.11 *Image tilt.* With the elevation level bubble centered, the image of a distant horizontal target line formed by the two optical systems shall be parallel within fifteen minutes of arc and both images shall be horizontal within one degree.

3.6.12 *Plumb travel.* With the range finder positioned as in 3.4.1 the range finder shall then be set at maximum depression, the sighting reticle projected into the left eyepiece, and starting with the line of sight through the boresight mark in coincidence with a plumb line, the line of sight shall not deviate from the plumb line by more than 0.20 mil when elevated through an excursion of 512 mils. In no case shall the summation of the maximum deviation to right and left of the plumb line exceed a total of 0.20 mil.

3.6.13 *Compensator elevation error.* When the ranging knob is actuated to cover the range from 480 to 4500 meters the elevation of the two target images shall not change with respect to each other by more than 0.10 mil.

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3.6.14 *Cross field errors.* From the median setting established in 3.6.2, and with the range finder at zero elevation using a 1500 meter (plus or minus 100 meters) target, establish coincidence of the target with the horizontal calibration knob at points 20 mils (plus or minus 2 mils) above, below, to the left and to the right of the center of the field (boresight cross). The median of five coincidence readings at each of the four points shall not differ from the median UOE reading obtained at the center of the field by more than 6 UOE.

3.6.15 *Ranging accuracy at changes in elevation and depression.* Range readings taken at plus 7 degrees and at maximum elevation (plus 14 degrees) and at maximum depression (minus 7 degrees) shall agree with the range readings obtained at zero elevation within 5 UOE when ranging on a target at 1500 meters, when all readings are taken at the boresight cross.

3.6.15.1 *Range scale accuracy.* With a target provided at 1500 meters ± 100 meters and with the range scale set to the range indicated, the horizontal calibration knob shall be adjusted for target range coincidence. When ranging on targets at the following ranges, the range scale shall agree with the target range within the tolerance specified below.

Target range in meters	Tolerance in meters
700	± 10
1000	± 20
1500 ± 100	0.0 (By adjustment)
2500	± 50
4000	± 50

3.6.15.2 *Synchro electrical zero.* The range data transmitter (15CX4a) shall be electrically zeroed within 15 minutes backlash at the 1500 meter range position.

3.6.15.3 *Range data output.* The output of the range data synchro transmitter (15CX4a), when energized by a 115 V, 400 cycle source and connected to a control receiver synchro shall cause the control receiver to produce the values within the tolerances as indicated in Table I.

TABLE I. *Shaft Rotation of Range Data Synchro Transmitter*

Range scale	Degrees from 1500 meter			Tolerance	
500	78°	41'	CCW	1°	28'
600	65°	37'	CCW	1°	0'
700	54°	33'	CCW		45'
800	45°	01'	CCW		23'
900	36°	35'	CCW		26'
1000	29°	20'	CCW		29'
1200	15°	59'	CCW		35'
* 1500		0		0°	20'
2000	20°	36'	CW		58'
2500	36°	35'	CW	1°	13'
3000	49°	39'	CW	1°	28'
3500	60°	41'	CW	1°	42'
4000	70°	15'	CW	1°	57'
4500	78°	41'	CW	2°	11'

* Synchro electrically zeroed at 1500 meters.

3.6.16 *Operational stop.* The operational mechanical stop of the range finder shall allow an excursion in range of at least 480 to 4500 meters.

3.6.17 *Elevation accuracy.* Starting with the range finder line of sight horizontal (zero elevation), counterclockwise rotation of the input coupling shall elevate the line of sight, clockwise rotation shall depress the line of sight, in conformance with the values in Table II.

TABLE II. *Elevation Accuracy*

Rev. of input coupling	Elevation in mils line of sight	Tolerance in mils line of sight
4	256.0	± 0.20
3	192.0	± 0.20
2	128.0	± 0.20
1	64.0	± 0.20
0	0	± 0.00
1	64.0	± 0.20
2	128.0	± 0.20

3.6.17.1 *Backlash in elevation.* The values recorded in rotating six revolutions clockwise shall not differ from the counterclockwise values by more than 0.1 mil.

3.6.18 *Cant correction.* With the range

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finder leveled as in 3.4.1 (d) the cant correction knob shall have an excursion of at least 2.75 revolutions each side of the detent. Movement of the sighting reticle resulting from angles of cant introduced by rotation of the cant control knob in each direction, and rotation in turns of the input shaft shall be in accordance with the values of Table III.

TABLE III. *Cant Correction*

(Reticle movement in mils)

	Rotation in turns of input shaft					
	1.0	3.5	6.0	8.5	11.0	13.5
0-REV.	0 ± 0.25	0 ± 0.25	0 ± 0.25	0 ± 0.25	0 ± 0.25	0 ± 0.25
1-REV.	0.1 ± 0.15	0.7 ± 0.20	1.3 ± 0.25	1.9 ± 0.3	2.6 ± 0.35	3.6 ± 0.4
2-REV.	0.15 ± 0.15	1.4 ± 0.20	2.6 ± 0.25	3.8 ± 0.3	5.0 ± 0.35	6.3 ± 0.4

3.6.18.1 *Backlash in cant correction.* With the input shaft at zero rotation, setting of the cant correction engagement in detent from either direction shall not cause a shift in the line of sight in excess of 0.05 mil.

3.6.19 *Base length.* The base length of the range finder measured optically with end housings in place shall be 73 inches plus or minus 1.0 mm.

3.6.20 *Controls.* All controls and knobs shall have sufficient movement to cover the entire range of their corresponding scales.

3.6.20.1 *Elevation boresight knob.* The elevation boresight knob shall have at least 5.5 mils travel for depression and 3.5 mils for elevation from the nominal position established in 3.4.2.

3.6.20.2 *Azimuth boresight knob.* The azimuth boresight knob shall have at least 4.0 mils deflection each side of the nominal position established in 3.4.2.

3.6.20.3 *Backlash in boresight knobs.* Backlash in the boresight knobs shall not exceed 0.05 mil in the line of sight as read on the knob scale.

3.6.20.4 *Boresight locks.* Actuation of the boresight locks shall not cause more than 0.05 mil shift in the line of sight, and not more than 0.05 mil rotation of the boresight knobs as read on the knob scale.

3.6.20.5 *Boresight lock retention.* When the boresight locks are in the locked position the boresight knobs shall not move when a torque of 12 pound inches is applied to the knobs.

3.6.21 *Operability.* When the range finder is exposed to the ambient temperatures indicated in 3.10.2, the maximum torque required for operation of the controls shall not exceed the values specified in Table IV.

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TABLE IV. *Operability*

Control	+60 to 90°F	−25°F	+150°F
Range knob	70 oz inch	16 lb inch	70 oz inch
Azimuth boresight	60 oz inch	10 lb inch	60 oz inch
Elevation boresight knob	60 oz inch	10 lb inch	60 oz inch
Halving knob	50 oz inch	8 lb inch	50 oz inch
ICS Knob	50 oz inch	8 lb inch	50 oz inch
Interpupillary knob	5 lb inch	8 lb inch	5 lb inch
Eyepiece diopter scales	30 oz inch	10 lb inch	30 oz inch
Cant correction knob	50 oz inch	10 lb inch	50 oz inch
Rheostat knob	30 oz inch	8 lb inch	30 oz inch
Light switch knob	m.o. *	m.o.	m.o.
Cant correction input coupling	30 oz inch	50 oz inch	50 oz inch
Elevation input coupling	6 lb inch	12 lb inch	12 lb inch
Boresight and calibration locks	m.o.	m.o.	m.o.
Transfer prism lever	m.o.	m.o.	m.o.
Filter lever	m.o.	m.o.	m.o.
Servo switch	m.o.	m.o.	m.o.
Horizontal calibration knob	30 oz inch	50 oz inch	30 oz inch
Vertical calibration knob	30 oz inch	50 oz inch	30 oz inch
Occluder lever	30 oz inch	70 oz inch	30 oz inch

* m.o. — mechanically operable.

3.7 Sealing. The range finder body and end housings shall show no evidence of leakage when subjected to an internal pressure of ten inches of mercury at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) for a period of twelve hours. Correction shall be made for any variation resulting from temperature and barometric changes.

3.8 Flushing. Upon satisfactory completion of the sealing requirements of 3.7, the range finder and the end housings shall be flushed with dry nitrogen. The flushing shall continue until the gaseous mixture bled from the range finder and each end housing have a dew point of at least as low as minus twenty five degrees Fahrenheit. The instruments shall be sealed with an internal pressure of dry nitrogen of between 0.20 and 0.50 inches of mercury.

3.9 Environmental requirements.

3.9.1 *Storage temperatures.* The range finder with the identified end housing attached, shall be exposed to ambient temperatures of minus 80 and plus 160 degrees Fahrenheit until thermal equilibrium is reached. After thermal equilibrium has been re-established at standard ambient temperature, the range finder and end housings shall show no evidence of physical failure such as cement separation, bonding failure, glass breakage, casting failure, etc., and shall meet the requirements of 3.10.

3.9.2 *Operational temperatures.* The range finder with the identified end housings attached shall be exposed to an ambient temperature of minus 40 degrees Fahrenheit until thermal equilibrium is reached, and inspected at that temperature to insure that all knobs, locks and operational adjustments are mechanically operable. The range finder with end housings attached shall then be exposed to an ambient temperature of minus 25 degrees Fahrenheit until thermal equilibrium is reached and then shall be inspected at this temperature for conformance with the requirements of 3.6.21 and Table IV. The range finder with end housings attached shall be exposed to an ambient temperature of plus 150 degrees Fahrenheit until thermal equilibrium is reached and inspected at this

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temperature for conformance to the requirements of 3.10.

3.10 Post-environmental check.

3.10.1 *Optical check.* Any range finder which has been subjected to any part of the environmental requirements of 3.9 shall be restored to the conditions of 3.4.1 (a) to (h) inclusive and shall comply with the following requirements after thermal equilibrium has been re-established at standard ambient temperature.

(a) The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.5 mil in deflection (azimuth) and 0.5 mil in elevation from the values in 3.4.5. The line of sight of the right telescope shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.3 mil. Coincidence of the target images shall be re-established by using the horizontal calibration knob only. Set the horizontal calibration knob to the new median reading of five horizontal coincidence readings of the target then, using the ICS knob, obtain a median from five ICS settings of the calibrating reticle. This median shall not differ by more than 5 UOE from median reading established in 3.6.2 prior to environmental exposure. (See 3.10.2.)

(b) The field shall not have increased from the values established in 3.6.14 by more than 2 UOE.

3.10.2 *Horizontal knob.* Subsequent to the environmental requirements of 3.9.1 or 3.9.2 horizontal coincidence of the range finder, using a target at 1500 meters plus or minus 100 meters, shall be within 30 UOE of the midpoint of total travel of the horizontal calibration knob. Should horizontal coincidence exceed 30 UOE the range finder will be acceptable providing the range finder is recalibrated to bring horizontal coincidence to within 10 UOE of the midpoint of total travel of the horizontal calibration knob. Following recalibration the range finder shall be rechecked and shall meet the following requirements: 3.6.4.1 (Dirt), 3.6.5 (Parallax), 3.7 (Sealing), 3.8 (Flushing).

3.10.2.1 *Coincident reticle setting.* Five reticle coincidence settings shall be made with the ICS knob. The median of the five readings shall read within 12 UOE of the zero position.

3.11 Electrical. Electrical checks may be conducted at any time prior to final acceptance of the range finder.

3.11.1 *High potential.* Insulation breakdown shall not occur between isolated circuits and between body or frame when 500 volts root means square (rms) is applied for a period of at least thirty seconds.

3.11.2 *Insulation resistance.* Following the high potential test, insulation resistance between isolated circuits and between body or frame shall not be less than 2 megohms.

3.11.3 *Continuity.* The electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to

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the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and List of Supplementary Quality Assurance Provisions, LSQAP-10511227. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table V titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE V. *Classification of Defects*

Use Inspection Level II in Table I with Sampling Plan in Table II of Standard MIL-STD-105

		Requirement	*Test procedure*
Critical: None defined.			
Major: AQL 1.0 Percent.			
101.	Sighting reticle	3.4.2	4.5.2
102.	Sighting reticle with filters	3.4.2.1	4.5.2.1
103.	Collimation	3.4.3	4.5.3
104.	Horizontal calibration knob	3.4.4.1	4.5.4
105.	Vertical calibration knob	3.4.4.2	4.5.5
106.	Horizontal calibration reticle	3.4.5.1	4.5.6
107.	Vertical calibrating reticle	3.4.5.2	4.5.7
108.	Post vibration	3.6	4.5.8.1
109.	Target image	3.6.1	4.5.9
110.	Coincidence setting reticle	3.6.2	4.5.10
111.	Vertical movement accuracy	3.6.3.1	4.5.11.1
112.	Horizontal movement accuracy	3.6.3.2	4.5.11.2
113.	Cleanliness (dirt)- post vibration	3.6.4.1	4.5.12
114.	Bright spots or pinholes	3.6.4.1	4.5.12
115.	Parallax sighting reticles	3.6.5.1	4.5.14
116.	Parallax calibrating reticle	3.6.5.2	4.5.15
117.	Parallax. Target image	3.6.5.3	4.5.16
118.	Interpupillary adjustment	3.6.6.1	4.5.17
119.	Eyepiece focus	3.6.6.2	4.5.18

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TABLE V. *Classification of Defects*—Continued

Use Inspection Level II in Table I with Sampling Plan in Table II of Standard MIL-STD-105

		Requirement	*Test procedure*
120.	Diopter setting	3.6.6.3	4.5.19
121.	Exit pupil	3.6.6.4	4.5.20
122.	Exit pupil coincidence	3.6.6.5	4.5.21
123.	Magnification	3.6.6.6	4.5.22
124.	Resolution	3.6.7	4.5.23
125.	Insufficient light transmission	3.6.8	4.5.24
126.	Sighting reticle measurement	3.6.9	4.5.25
127.	Parallelism, sighting reticle and image	3.6.10	4.5.26
128.	Image tilt	3.6.11	4.5.27
129.	Plumb travel	3.6.12	4.5.28
130.	Compensator elevation	3.6.13	4.5.29
131.	Cross field error	3.6.14	4.5.30
132.	Ranging accuracy	3.6.15	4.5.31
133.	Range scale	3.6.15.1	4.5.32
134.	Synchro electrical zero	3.6.15.2	4.5.33
135.	Range data output	3.6.15.3	4.5.34
136.	Operational stop	3.6.16	4.5.35
137.	Elevation accuracy	3.6.17	4.5.36
138.	Backlash elevation	3.6.17.1	4.5.36
139.	Cant correction	3.6.18	4.5.37
140.	Backlash cant correction	3.6.18.1	4.5.38
141.	Base length	3.6.19	4.5.39
142.	Elevation boresight knob travel	3.6.20.1	4.5.10.1
143.	Azimuth boresight knob travel	3.6.20.2	4.5.40.2
144.	Backlash boresight knobs	3.6.20.3	4.5.40.3
145.	Boresight locks	3.6.20.4	4.5.40.4
146.	Retention boresight locks	3.6.20.5	4.5.40.5
147.	Operability, torque	3.6.21	4.5.41
148.	Sealing	3.7	4.6
149.	Dew point flushing	3.8	4.7
150.	High potential	3.11.1	4.10
151.	Insulation resistance	3.11.2	4.10.1
152.	Continuity	3.11.3	4.10.2
Minor:	AQL 4.0 Percent.		
201.	Paint and burrs on locating and bearing surfaces	3.3.2(f)	Visual
202.	Locking compound on exterior screw heads	3.3.2(f)	Visual

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL-STD-105.

4.3 Control tests. One range finder shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	Requirement	Test procedure
Materials	3.1	Appropriate Specification
Fabrication	3.2	Visual — As Per Drawing
General Specifications	3.3	Visual — MIL-F-13926 and MIL-O-13830
Storage Temperatures	3.9.1	4.8
Operational Temperature	3.9.2	4.8.1

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4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment," and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Inspection equipment design requirements.*

4.4.2.2.1 *Equipment to test complete range finder (Dwg. F10511227).* This equipment shall contain the following requirements and associated accessories.

(a) The equipment shall have a suitable means of mounting the range finder, by means of the mounting bearings, in the normal operating position. The equipment shall be so constructed as to permit cross-leveling within 10 seconds, and the horizontal mounting surface of the device shall be horizontal and parallel with respect to a datum line established (theoretically) by the mechanical centerline of the range finder within 25 seconds.

(b) An appropriate target or targets, with acceptance and rejection limitation markings or pips marked thereon, utilized for measuring the reticle patterns, shall be plumb and in line vertically and horizontally with the datum line within 25 seconds. The target or targets shall be so constructed as to be able to simulate target distances of from 480 meters to 4500 meters, The left line of sight shall be 90 degrees with respect to the datum line within 30 seconds. The target lines of sight shall be parallel to each other in deflection within 1.5 seconds and parallel to each other in elevation within 10 seconds. There shall be sufficient locating surfaces for any other required auxiliary testing equipment use.

(c) There shall be equipment capable of supplying electrical power of the current, voltages and frequencies necessary to perform required inspection tests.

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(d) The equipment shall be capable of elevating and depressing within the tolerances specified in 3.6.17.

(e) The equipment shall have a means of checking range data output as follows:

Degrees from 1500 meters	Tolerance
—78° 41'	8'8"
—65° 37'	6'0"
—54° 33'	4'50"
—45° 1'	2'30"
—36° 33'	3'0"
—15° 59'	3'50"
0	0(±0°2' Elec Zero at 1500)
+20° 35'	6'20"
+36° 33'	7'30"
+49° 39'	9'20"
+60° 40'	10'20"
+70° 14'	12'10"
+78° 41'	13'10"
+83° 18'	14'0"
+86° 13'	15'0"

4.4.2.2.2 *Rigidity and stability.* The calibrating and testing equipment shall have ease of adaptability and set up of calibrating equipment, for frequent necessary inspection surveillance checks.

4.4.2.3 *Equipment for vibration test.* This special test equipment shall be capable of producing vertical motion at an amplitude of 1/16 inch (1/8 inch total movement) at a frequency of 30 cycles per second, and must meet the tolerance specified in Specification MIL–F–13926.

4.4.2.4 *Equipment to test torque.* A torque wrench tester (Std. Meas. Equip.) having a range from zero to twenty five inch pounds, the accuracy of which shall be plus or minus 1/4 inch pounds tolerance, shall be used to determine the requirements of 3.6.21. This equipment shall be equipped with various special adapters to accommodate testing of the following: boresight knobs, range knob, halving knob, ICS knob, interpupillary knob, cant corrector knob, rheostat knob and eyepiece diopter scales.

4.5 Test methods and procedures.

4.5.1 *Orientation.* The requirements for orientation 3.4.1 (a) to (l) shall be met with the range finder mounted on the Special Test Equipment. The mounting surfaces of the right bearing support and the elevation assembly on the left shall be in a plane parallel to the horizontal line of sight through the mounting bearings. The line of sight through the range finder shall be directed to a calibrated target, positioned at the specified distance, appropriate for measuring the accuracy requirements of the range finder. The target shall be positioned perpendicular to the range finder mounting surfaces.

4.5.2 *Sighting reticle alignment.* This inspection test shall be conducted with the range finder in position as in 4.5.1 with the sighting reticle projected into the left telescope, the center of the boresight cross shall coincide with the boresight cross of the right telescope within the tolerance specified in 3.4.2 without filters.

4.5.2.1 *Filters.* The introduction of filters into the optical systems shall not cause the reticle alignment to vary or change more than the tolerance specified in 3.4.2.1.

4.5.3 *Collimation.* Collimation inspection check shall be conducted with the aid of an auxiliary combining telescope (Special Test Equipment) positioned at the eyepiece of the range finder. Divergence and dipvergence shall be measured on the telescope reticle. "Divergence" is displacement in azimuth of two images and "dipvergence" the displacement in elevation between two images. With the IPD knob set at 65 mm, sight through the telescope and measure the distance from the left target image to the right target image in azimuth (divergence) on the telescope scale, the measurement shall not exceed the tolerance specified in 3.4.3. Measure the distance from the left to the right in elevation (dipvergence) and record readings. Readings at both extremes of the IPD knob shall be made and also reverse readings for

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backlash effect on dipvergence and divergence. See Figure 2 for examples of dipvergence and divergence.

4.5.4 *Horizontal calibration knob.* This inspection test shall be performed to determine the center of rotation within the tolerance specified in 3.4.4.1. Rotate knob to one extreme and mark across scale and index. Rotate to other extreme and count turns on scale. Divide turns by two to determine center of rotation.

4.5.5 *Vertical knob.* This inspection test is to determine the center of rotation within the tolerance specified in 3.4.4.2 and shall be performed as in 4.5.4 above.

4.5.6 *Horizontal setting, reticle accuracy.* This inspection shall consist of a series of readings made with the ICS knob and shall meet the requirements of 3.4.5.1.

4.5.7 *Vertical setting, reticle accuracy.* This inspection shall determine vertical coincidence position of the calibrating reticle, with respect to halving knob as specified in 3.4.5.2.

4.5.8 *Vibration.* The complete range finder shall be mounted in its normal operating position on the Special Test Equipment and vibrated in accordance with 3.5.

4.5.8.1 *Post vibration.* This inspection, to insure that no physical failure of mechanical or optical components has occurred or that certain changes in the line of sight or collimation have not taken place in accordance with 3.6 (a) and 3.6 (b).

4.5.9 *Target image.* The horizontal coincidence of the target image shall not change more than the tolerance specified in 3.6.1 when measured against a calibrated external target. A series of readings shall be made to determine the UOE, using the horizontal calibration knob.

4.5.10 *Coincident reticle setting.* A series of readings shall be made to determine a new median using the ICS knob to set the calibrating reticle. There shall not be a difference of more than the specified UOE in 3.6.2 and the original setting established in 3.4.5.1.

4.5.11 *Coincident reticle's position.* The range finder shall be reset to the requirement of 3.6.3 and the following inspection tests shall be made.

4.5.11.1 *Vertical movement accuracy.* Rotate the vertical calibration knob to the maximum counterclockwise position (CCW) and then, by using the halving knob, return the reticle to its original position. The median of five deflection UOE readings of the calibrating reticle, using the ICS knob, shall not vary by more than 10 UOE from the median of 5 deflection readings obtained when the vertical calibration knob is rotated to maximum clockwise (CW) position and the reticle returned to its original position by using the halving knob as specified in 3.6.3.1.

4.5.11.2 *Horizontal movement accuracy.* Rotate the horizontal calibration knob from the median setting 3.6.3.2 to plus 40 and minus 40 UOE. The range at each extreme shall be equivalent to the corresponding change in the number of UOE on the knob scale within tolerance specified in 3.6.3.2. In addition there shall be 5 UOE allowed to cover errors in the position of the horizontal calibrating wedge with respect to the ranging plane and its angullar tolerance specified in 3.6.3.1.

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For computing the change in range that corresponds to the change in UOE, use the following formula:

$$\text{Change in Range} = \frac{3.3\ nR^2}{1 \pm .0033\ nR}$$

Where:

R = Range in Meters / 1000

n = Number of UOE

+ = Decreasing the Range

Example

(a) For Increasing Range from 1500 Meters

$$\text{Change in Range (Meters)} = \frac{3.3\ nR^2}{1 - .003\ nR} =$$

$$\frac{(3.3)\ (40)\ (2.25E)}{1 - .003\ (40)\ (1.5)} = 370.3$$

(b) For Decreasing Range from 1500 Meters

$$\text{Change in Range (Meters)} = \frac{3.3\ nR^2}{1 + .0033\ nR} =$$

$$\frac{(3.3)\ (40)\ (2.25)}{1 + .0033\ (40)\ (1.5)} = 247.9$$

4.5.12 *Dirt.* There shall be no excess dirt visible in the areas as specified in 3.6.4.1. Any visible dirt or digs on or in the optical surfaces shall not exceed the tolerance and cleanliness requirements specified in Specification MIL–O–13830. There shall not be any other imperfections that will impair the observation, ranging or image quality.

4.5.13 *Bright spots.* This is a visual inspection and the bright spots (pin holes) that are visible through the opaque or coated areas shall not exceed the size and number of digs for that component, compared to the dig standards specified in Specification MIL O–13830.

4.5.14 *Parallax between sighting reticle and target images.* This is a visual inspection test utilizing a target in conformance with 3.6.5.1. Readings shall be made on the boresight scale and the amount of parallax present shall not exceed the amount specified.

4.5.15 *Parallax of the calibrating reticles.* This inspection test shall be conducted with the aid of an auxiliary collimator and visual means. The auxiliary collimator shall be positioned at the range finder eyepiece. A series of 5 readings, using the ICS knob, shall be made with the eye in the positions specified in 3.6.5.2. The amount of parallax present shall not exceed the tolerance specified.

4.5.16 *Parallax of the target images.* The auxiliary collimator used in 4.5.15 shall be utilized for this inspection. A target as specified in 3.6.5.3 or equivalent shall be used. A series of 5 readings shall be made, with the eye positioned as specified. The amount of parallax shall not exceed the tolerance specified when read on the horizontal calibration knob.

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4.5.17 *Interpupillary adjustment.* This inspection shall be made at the 65 mm and two extreme settings of the IPD scale. This test shall be conducted with a combining telescope positioned at the range finder eyepiece and the target images emerging from the eyepiece into the test equipment's single eyepiece combined for accurate measurement. This is accomplished by having the opaque dots on the objective lenses (Drawing A7632698) in focus and bringing them into coincidence in the test equipment and reading the microscope objective assembly scale on the test equipment. The IPD value shall be recorded and shall be within the specified tolerance, as required by 3.6.6.1.

4.5.18 *Eyepiece focus.* The individual eyepieces shall be inspected manually and visually, the eyepieces shall travel from a zero to a full plus three diopters and a minus three diopters as read against the index line, and shall meet the requirement in 3.6.6.2.

4.5.19 *Diopter setting.* With the aid of a dioptometer this inspection test shall be conducted to insure that the zero on the diopter scale is within the specified tolerance of the index required in 3.6.6.3.

4.5.20 *Exit pupil diameter.* Inspection of the exit pupil diameter shall be conducted with standard measuring equipment. The procedure is to bring the exit pupil into sharp focus and read the diameter. The tolerance must meet the requirement of 3.6.6.4 for each eyepiece.

4.5.21 *Exit pupil coincidence.* This inspection test shall be conducted with special test equipment. Each exit pupil shall be circular and be in coincidence with one another within the specified tolerance in 3.6.6.5.

4.5.22 *Magnification.* The inspection for Magnification shall be conducted in accordance with Specification MIL-O-13830 and shall meet the requirements of 3.6.6.6.

4.5.23 *Resolution.* Inspection of resolution shall be in accordance with Specification MIL-O-13830 and must be within the tolerance specified in 3.6.7.

4.5.24 *Light transmission.* Light transmission on the right and left optical systems shall be inspected with special test equipment to determine the value as required by 3.6.8. This is accomplished by placing a light source (Min. 15 watt bulb) in front of each end housing assembly and the special test equipment at the right and left eyepiece. An average for series of readings shall be computed. The difference between the right and left systems shall not exceed the specified requirement.

4.5.25 *Sighting reticle measurement.* This inspection shall be conducted in accordance with Specification MIL-O-13830. The angle subtended by the total horizontal lines and spaces of the right and left sighting reticle shall meet the requirements specified in 3.6.9.

4.5.26 *Parallelism of sighting reticle and image.* This inspection shall be conducted with the aid of an auxiliary telescope on special test equipment. By viewing the 40 mil horizontal reticle line, measure the tilt to the target image on the graduation at either end of the target image scale. Tilt of reticle must be within the tolerance as specified in requirement 3.6.10. Forty three minutes of arc is equivalent to an angle subtended by a 0.50 mil block at one extremity of the reticle line when the other extremity of the reticle line is in coincidence with the target line as shown in Fig. 1.

4.5.27 *Image tilt.* This inspection test shall be conducted with the aid of an auxiliary telescope (special testing equipment) designed to project the combined target images of the right and left optical systems on to a reticle pattern that can measure image tilt. This is accomplished by leveling the auxiliary telescope, then placing it at the range

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finder eyepiece. Rotate the range finder until the right and left target images are on the auxiliary telescope measuring reticle and the elevation level bubble is centered and check the tilt. Continue the range finder in the same direction until images are beyond view, reverse direction until the level bubble is centered and check the tilt. All tilt readings must meet the requirements specified in 3.6.11 and illustrated in Fig. 2.

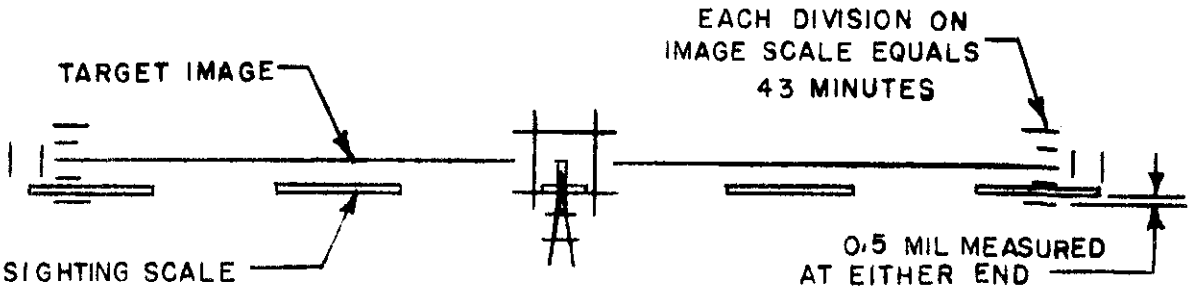


FIGURE 1.

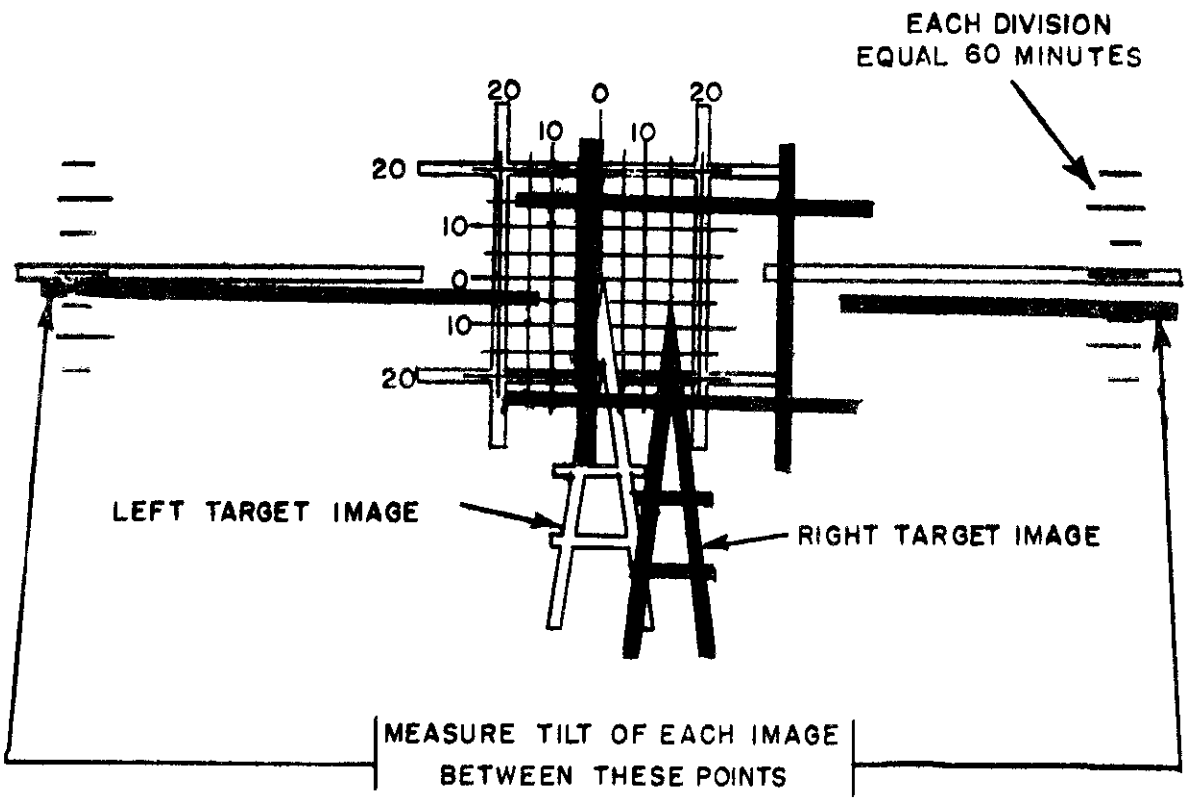


FIGURE 2.

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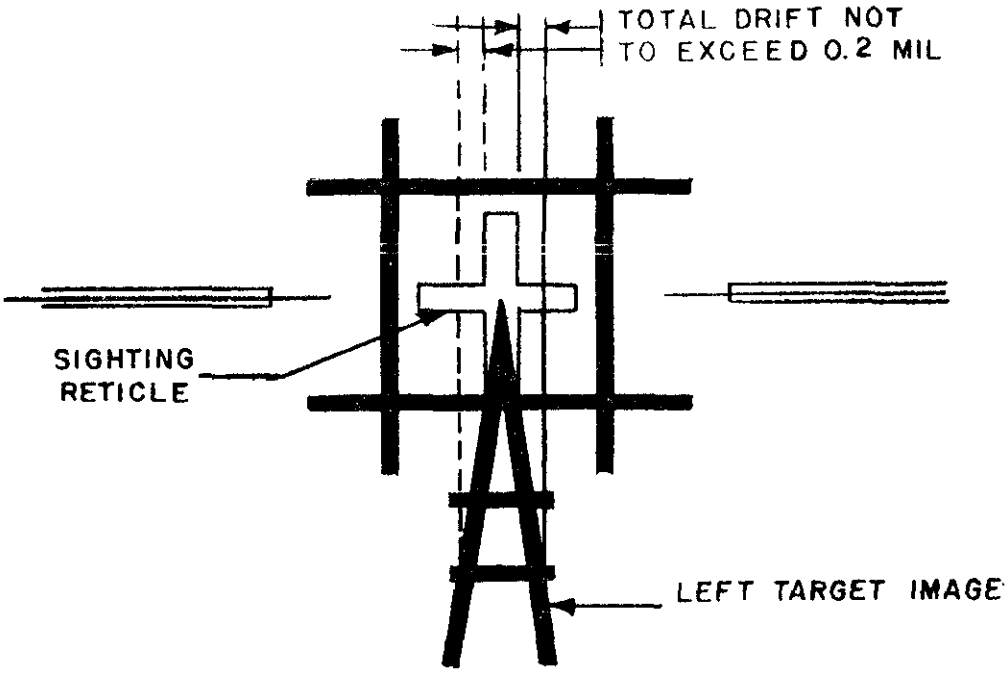


FIGURE 3.

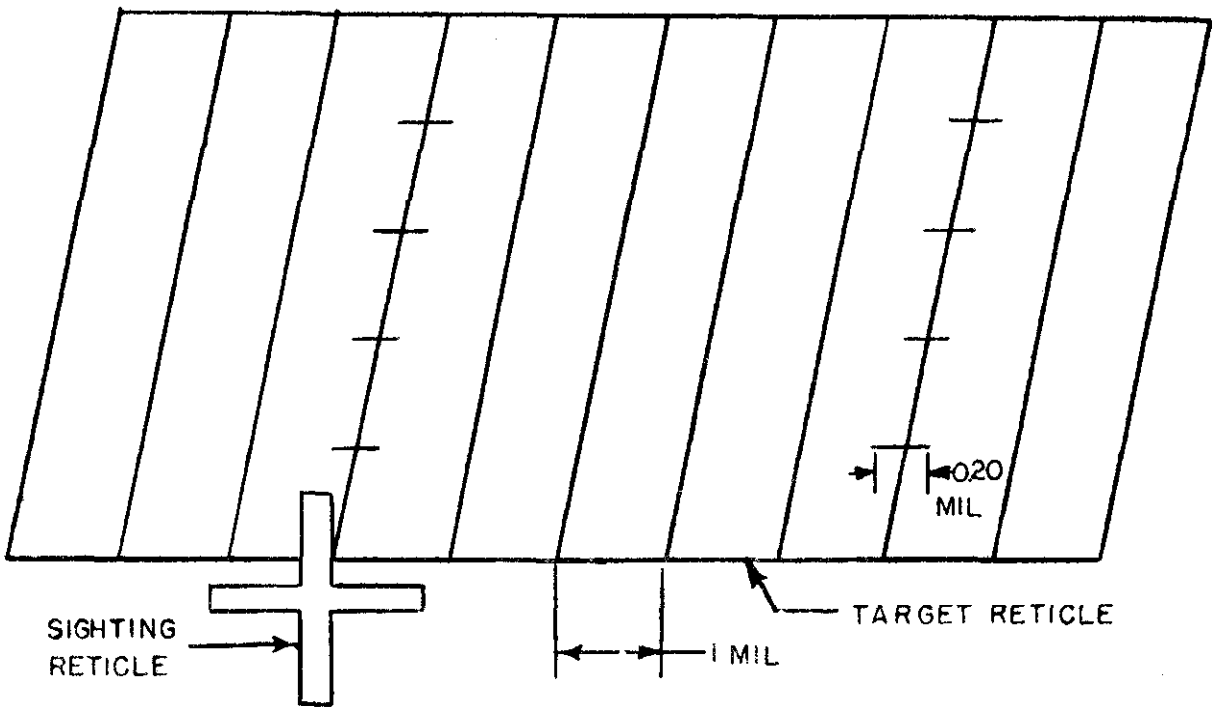


FIGURE 4.

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4.5.28 *Plumb travel.* The inspection test for plumb travel will be conducted with the aid of an auxiliary telescope while the range finder is in position as described in 3.4.1. Place the auxiliary telescope in position in front of the left eyepiece. Align the sighting reticle with the left target image in azimuth through the left eyepiece. Depress the range finder to 12 degrees maximum depression and with the auxiliary telescope measure the distance that the sighting reticle has drifted or moved in azimuth. Elevate the range finder to its maximum of 15 degrees and measure azimuth drift or movement of the sighting reticle again. The total drift of the sighting reticle shall not exceed specified requirements in 3.6.12 and lilustrated in Figure 3.

4.5.29 *Compensator elevation error.* This inspection test shall be conducted utilizing the auxiliary telescope (special testing equipment) used in 4.5.27. Locate the auxiliary telescope at the eyepiece of the range finder. Black out the left side optical system for identification of targets. After identification is made, open left optical system then check the plane of target images. Traverse the full excursion of the range from 480 to 4500 meters while viewing the two target images against the auxiliary telescope reticle. This test shall meet the requirements of 3.6.13.

4.5.30 *Cross field errors.* This test shall be conducted with five targets positioned at 1500 meters plus or minus 100 meters at zero elevation. One target shall be positioned at the center of the field (a line of sight through the reticle boresight mark) with the remaining four targets positioned at the specified number of mils (see 3.6.14) from the center of the field. Set the range finder range scale to the range chosen and sight on the target in the center of the field. The target at the center of the field shall be in coincidence when the horizontal calibration and ICS knobs are set to their established median positions. A series of five horizontal coincidence settings shall be taken on each target in turn by means of the horizontal calibration knob. The difference between the median of the center target and the medians of each group of series of five settings shall not exceed the tolerance specified in 3.6.14 as measured on the scale of the horizontal calibration knob.

4.5.31 *Ranging accuracy at changes in elevation and depression.* The inspection for range accuracy shall be conducted with the range finder mounted on special test equipment. With the line of sight to the target horizontal, set the range scale to 1500 meters. Take a series of five readings, with the ICS knob and establish a median reading. Elevate the entire range finder so that the line of sight is to plus 7 degrees and again to plus 14 degrees above the horizontal. Take a series of five readings at each position (7° and 14°) with the ICS knob and establish a median reading. The median readings shall meet the requirements of 3.6.15.

4.5.32 *Range scale accuracy.* The inspection for range scale accuracy shall be conducted with the range finder mounted on special test equipment. Set the range scale at 1000 meters. Set the special test equipment, range compensator to set target at 1000 meters. Take a series of five readings with the ICS knob and set the ICS median. Reset target at 520 meters and range with range knob for coincidence of target image. Make five readings and note the median. Repeat procedure for 700, 1000, 1500, 2400 and 4000 meters range. The median readings in meters shall meet the tolerances specified in 3.6.15.1. The following formula shall be used for increasing or decreasing ranges. One UOE in meters is equal to $\frac{3.3\ (R)^2}{1000}$ where R is the range in thousands of meters.

4.5.33 *Synchro electrical zero.* The inspection for electrical zero shall be conducted with standard measuring equipment and

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special test equipment. With the multimeter AC volts test equipment connected to the range finder receptacle terminals A and C, connect the range finder to the special test equipment terminals S1, S2, S3, R1 and R2 respectively on the special test equipment. Terminals R1 and R2 are also connected to 110 volt, 400-CPS on the special test equipment. Range finder terminals F and G are connected to the 24 volt DC power on the special test equipment. Turn the volts test instrument scale selector to 100, turn power switch to ON position on the special test equipment and the 24 volts DC power to ON. Set DC voltage to 24, 400 cycles per second power to ON and 400 CPS voltage to 110. Turn range knob on the range finder from fully clockwise (500 meters) to exactly 1500 meters, adjust the voltmeter scale selector until the minimum readable voltage is obtained, and record. Turn the range knob to move the range scale to 5000 meters and slowly return to the 1500 meter point. Read the voltage on the voltmeter (change scales to obtain a reading if necessary) and subtract the first voltage obtained from the second voltage obtained, multiply this result by 4. The figure obtained is the backlash, expressed in minutes of rotation, of the synchro shaft. This figure must not exceed the tolerance specified in 3.6.15.2.

Example: 1st Voltage Reading	=	0.7 volts
2nd Voltage Reading	=	3.0 volts
2nd Voltage	=	3.0
1st Voltage	=	0.7
		2.3 Volts Difference
		X4 Minutes Per Volt
Total		9.2 Minutes of Backlash

4.5.34 *Range data output.* This inspection test will utilize the special test equipment used in 4.5.33 connected to the range finder. The voltmeter shall be removed for this inspection and all power and voltage conditions shall be the same, with the range scale of the range finder set at 1500 meters. Set angular control dials (inner and outer) to exact zero, rotate both inner and outer dials simultaneously until a null point on the null meter of the special test equipment is obtained. This has established an electrical zero between the range finder synchro transmitter 15CX4a and the special test equipment synchro receiver 15CT4a. Rotate the range knob of the range finder to move the range scale to 1200 meters and coincide the range scale and index without overtravel. Rotate the inner scale on the test equipment (do not move outer dial, this is index point) until a null point is again obtained on the null meter, read the rotation in degrees and minutes, i.e., 15 degrees 29 minutes, plus or minus 35 minutes. Continue to inspect the range data output at each range listed in Table I, between 1500 and 500 meters, then reset range scale at 1500 meters and inspect each range listed from 1500 meters to 4500 meters, all readings should be recorded and they shall meet the tolerance specified in Table I and required by 3.6.15.3.

4.5.35 *Operational stops.* The inspection of the operational stops shall be a visual and manual check. This check should be made prior to installing the range finder in the special test equipment. Rotation of the range knob shall travel the range scale from one extreme (480 meters) to the other extreme (4500 meters) without hesitation, drag or jumping and the scale shall at least coincide with the index at the low range

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(480 meters), and go slightly past the index at the high range (4500 meters) as required by 3.6.16.

4.5.36 *Elevation accuracy and backlash.* With the range finder installed on the special test equipment and adjusted to a horizontal zero line of sight determined by using a precision level of 10 seconds or less (standard measuring equipment), and the left end housing removed, a series of inspection readings for elevation and depression shall be made. Set the sighting reticle exactly on the target image by using the azimuth and elevation boresight knobs. Lock the elevation knob and slip the scale to (4) (this is the indexing point). Attach a dial, or disc (special test equipment) graduated in degrees from 0 to 100, to the input coupling of the range finder, and using and improvised index, set the dial and index at zero degrees. Set a gunners quadrant at plus 64 mils and place it on the level pad in a position that will center the level bubble when the target is elevated and leveled at plus 64 mils. Rotate the manual elevation input knob to elevate the range finder. Make one complete turn on the graduated dial, stopping at zero without overtravel. Observe the position of sighting reticle and the target. Unlock the elevation boresight knob scale. Repeat this operation for plus 128 mils, 192 mils, and 256 mils elevation. After the 256 mil elevation reading has been made, elevate the range finder one turn beyond the 256 mil angle and then reverse the direction. Observe the rotation of dial and stopping without overtravel in this reverse direction make a reading at the 256 mil angle and record. Make decreasing angle backlash readings at 192 mils, 128 mils, 64 mils, and check backlash at zero. Continue to depress the range finder to obtain the minus 64 mil and the minus 128 mil readings. Repeat the inspection for backlash at 128 mils that was made at plus 256 mils. Increasing elevation, made readings at minus 64 mils and zero to determine backlash. All elevation readings and depression readings must meet the requirements of 3.6.17, 3.6.17.1 and Table II.

4.5.37 *Cant correction.* This inspection operation shall be conducted with the range finder mounted as specified in 4.5.1 and 3.4.1 (c) with an auxiliary collimator (special test equipment) mounted in target line of sight at the right end housing window on the leveling table of the test equipment. Set the cant correction knob on the range finder to detent, then rotate the splined input shaft (located on front of main housing) fully counterclockwise. Using the boresight knobs, set the sighting reticle on a reference point on the target reticle in the auxiliary collimator, (see Fig. 4). Mark the splined shaft to identify rotation made. Rotate the splined shaft clockwise 1, 3.5, 6, 8.5, 11 and 13.5 turns, respectively. The values in Table III shall be met. Return the splined input shaft fully counterclockwise, then rotate the cant correction knob one complete turn clockwise. Check the mark on the splined shaft and rotate the same number of revolutions as previous. Values shall meet requirements in Table III for one revolution on the cant correction knob. Repeat this procedure for two revolutions of the cant correction knob clockwise, then completely repeat the procedure for opposite rotation of the cant correction knob. When reading the position of the sighting reticle on the target reticle as the various turns are made, all readings shall be made from the starting position of the sighting reticle as it moves in azimuth. All readings shall meet the requirements of 3.6.18 within the tolerances of Table III.

4.5.38 *Backlash in cant correction.* This inspection shall be conducted with the aid of an auxiliary telescope at the eyepiece of the range finder. Visually check the movement of the sighting reticle while rotating the cant correction knob in either direction and back to detent. There shall be no shift in the line of sight in excess of the tolerance specified in 3.6.3.1. Check the backlash in the cant correction level mechanism. The

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bubble in the level shall be centered at zero or middle detent. The bubble should be centered within the width of a graduation line on the level vial. Several checks should be made for repeatability.

4.5.39 *Base length.* This inspection operation shall be conducted with the aid of the auxiliary test telescope used in 4.5.17. Position the auxiliary vertical wire length measuring adapters to the special test equipment in front of the window of the end box assembly. Illuminate with sufficient light for sharp focusing. Set the interpupillary knob on the range finder at 65 mm. Set the microscope objective millimeter assembly on the auxiliary test telescope to 65 mm. Position the auxiliary test telescope in front of the range finder eyepiece. View through the telescope to observe the vertical wires and focus for the sharpest image. If both wires are visible cover one eyepiece of the range finder at a time, and distinguish the right and left wires. Make sure the wires appear to the right and left as the right and left eyepiece is covered. If they appear opposite or converge, the base length is short. This condition is not acceptable. If the wires appear spread and they do not converge adjust the microscope objective assembly on the auxiliary telescope until the wire images are in coincidence or appear at one wire. The IPD value shall be recorded and subtracted from the value recorded in 4.5.17 for the median IPD setting (65 mm). The result shall be used as the "D" factor in the following formula.

$$X = D - X - 9.5 \text{ mm.}$$

X = Base length error in mm.

D = Difference in mm between dots and wires.

A = Divergence in minutes for range finder setting of 65 mm.

9.5 = Power (magnification) of range finder.

Convert "X" to inches by multiplying by 0.0394. The value of "X" shall not exceed the tolerance specified in 3.6.19.

4.5.40 *Controls.* A visual manual inspection of all controls and knobs shall be made to insure that sufficient movement or rotation is available, to meet the individual requirement as specified by 3.6.20.

4.5.40.1 *Elevation boresight knob.* This inspection shall be visual and manual. With the range finder level on the special test equipment and zero line of sight in elevation established, the boresight knob should be positioned at the Figure (4) on the knob scale. This is the nominal position. Rotate knob to depress sighting reticle and note the amount of travel. Rotation shall be 5.5 full mils depression. Reverse the direction and count the mil travel in elevation. Rotation shall be a full 9.0 mils to meet the requirements specified in 3.6.20.1.

4.5.40.2 *Azimuth boresight knob.* This inspection shall be visual and manual. The range finder shall be in the same condition as 4.5.40.1, except the azimuth boresight knob shall be positioned at the Figure (3) on the knob scale. This is the nominal position, and shall be at zero deflection on the target reticle. Rotate the boresight knob for right or left deflection and note the amount of travel in deflection from the nominal point. Rotation must meet the 4.0 mils travel required in 3.6.20.2. Reverse direction and count the mil travel in deflection. Rotation must be a full 8.0 mils to meet specified requirements.

4.5.40.3 *Backlash in boresight knobs.* The inspection for backlash shall be conducted with the aid of an auxiliary telescope posi-

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tioned in front of the eyepiece. Sighting through the auxiliary telescope, rotate the elevation boresight knob clockwise to the knob stop, then counterclockwise until the sighting reticle is centered on the target image at a point to make a positive reading. Read the knob scale for the setting made and record, continue in the counterclockwise direction until the opposite knob stop is reached, then turn the knob in a clockwise direction and position the sighting reticle at exactly the same point on the target image without overtravel. Record the reading on the knob scale. The difference between the two readings will be the amount of backlash present and must not exceed the tolerance specified in the requirement of 3.6.20.3. The above procedure shall be repeated for inspection of the backlash in the azimuth knob and shall meet the same requirements.

4.5.40.4 *Boresight locks.* With the aid of an auxiliary telescope, position the sighting reticle in a position on the target reticle to a point that, when the azimuth and elevation boresight locks are normally locked and unlocked, and movement or shift of the sighting reticle could be detected and the amount of movement can be measured on the respective knob scale. The tolerance must meet the requirements as specified in 3.6.20.4.

4.5.40.5 *Boresight lock retention.* With the correct torque adapter attached to the torque wrench and securely attached to the boresight knob, and the locking lever in the full locked position, the knob shall not rotate before the required inch pounds torque is reached in either direction as specified in 3.6.20.5.

4.5.41 *Operability.* When the range finder is exposed to the various temperatures in Table IV, all torque and manual requirements shall be met for the ambient temperatures indicated. All torque tests shall be accomplished to the correct adapter and all readings recorded. The recorded readings shall be within the tolerances specified in 3.6.21 and 3.9.2.

4.6 Sealing. The range finder and each end housing shall be inspected individually on special pressure testing equipment. The end housings can be removed from the range finder or left attached to the main housing while the tests are in progress. These tests shall be performed after the range finder has been removed from the range finder checking equipment. Attach the range finder main housing to the pressure test fixture at one cycling valve and the source of dry nitrogen gas at the other cycling valve. This procedure shall be followed for testing the end housing sealing. Fill the range finder and the end housings to required internal pressure specified in 3.7 for the prescribed period of time and record gage readings at various intervals. The range finder and each end housing shall meet the requirements in 3.7.

4.7 Flushing. This inspection procedure should be accomplished by the operator at the time the flushing is being performed. The dew point reading shall be made at this time to reduce excess use of nitrogen gas. This is done as the dry nitrogen gas is allowed to escape through the dew point checker. The moisture content of the gas is recorded. When the moisture content as specified has been reached, the range finder and end housing shall be sealed off with the required internal pressure as specified in 3.8.

Note: Caution shall be exercised to secure all knobs to prevent turning or rotating which allows gas to leak past shafting seals.

4.8 Storage temperature. A visual inspection of the complete range finder shall be made after being exposed to temperatures specified in 3.9.1. When thermal equilibrium has been established there shall be no evidence of physical failure, optically or mechanically, and there shall be no moisture present on optical surfaces visible to the unaided eye.

4.8.1 *Operational temperatures.* The inspection at operational temperatures shall

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be accomplished in conjunction with 4.5.41 (torque tests) for all temperatures required by 3.6.21 and Table IV.

4.9 Post environmental optical check. Subsequent to the environmental requirements of 3.9, the range finder shall be mounted on special test equipment and adjusted as specified in 4.5.1, and shall be set-up as required in 3.4.1. A complete inspection shall be made as prescribed by 4.5 to 4.7 inclusive. The range finder shall meet all requirements specified in 3.10.

4.10 Eelctrical, high potential. The electrical inspection test shall be conducted with special test equipment for circuit testing. Connect the test equipment to the range finder in the same manner as 4.5.33. Switch the test equipment to ON. Turn continuity test selector to breakdown position on the stator, rotate the power stator to increase voltage to specified value in 3.11.1 for the required length of time, and slowly reduce voltage to zero volts. Turn selector to ROTOR and repeat voltage test. If insulation breakdown should occur it will register on the test equipment.

4.10.1 *Insulation resistance.* This test shall be conducted with the range finder connected to the test equipment as in 4.10. Turn continuity selector to OFF position and rotate power stator to required voltage in 3.11.1, turn the lamp selector switch to STATOR through S1–S3, S1–52 and S2–53 positions. The insulation resistance in megohms will register on the megohm meter. Repeat operation for the rotor for R1–R2. All values shall meet the specified requirements of 3.11.2.

4.10.2 *Continuity.* The inspection of the various electrical circuits within the range finder for continuity shall be made in conjunction with 4.10.1. Position continuity selector switch to CONTINUITY TEST position. Turn lamp selector switch to ROTOR and STATOR positions and check for GOOD, OPEN or SHORT on continuity test meter. Turn selector switch to LAMPS for DIM and BRIGHT and the range finder reticle switch to corresponding position. Continuity meter must read GOOD. Turn off all switches and disconnect the range finder from the test equipment.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P10511227. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. This range finder is intended for installation in the 120 mm Gun, Heavy Tank: M103A1. It is commander operated and is primarily intended to determine ranges between 500 and 5000 meters and furnish range data to an electromechanical super-elevation computing system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing (5.1).

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

Custodian:
Army—MU

Preparing activity:
Army—MU

Project No. 1240–A392Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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